EXP.NO: 3	Write a Servlet program that prints "Hello, World!" when accessed through a browser.
DATE:20/02/2025	

Aim :

To develop a Servlet-based Online Quiz Application that accepts user details, displays quiz questions, processes responses, and shows the final score when accessed through a browser.

Algorithm :

1. Input:
   - The user accesses the quiz application through a browser
   - The user enters details (Name, etc.) and selects answers for quiz questions

2. Servlet Program Flow:
1. Create a Servlet Class
   - Define a Java class that extends HttpServlet
   - Override doGet() and doPost() methods

2. Handle HTTP Request
   - In doPost() method:
     - Retrieve user inputs using request.getParameter()
     - Process quiz answers
     - Calculate score

3. Generate Response
   - Set response type as text/html
   - Use PrintWriter to display:
     - User name
     - Quiz result (score)

4. Compile the Servlet
   - Use javac to compile .java file

5. Deploy the Servlet
   - Place .class file in /WEB-INF/classes
   - Configure using annotation or web.xml

6. Run Application
   - Start Apache Tomcat Server
   - Open browser:

CODE:

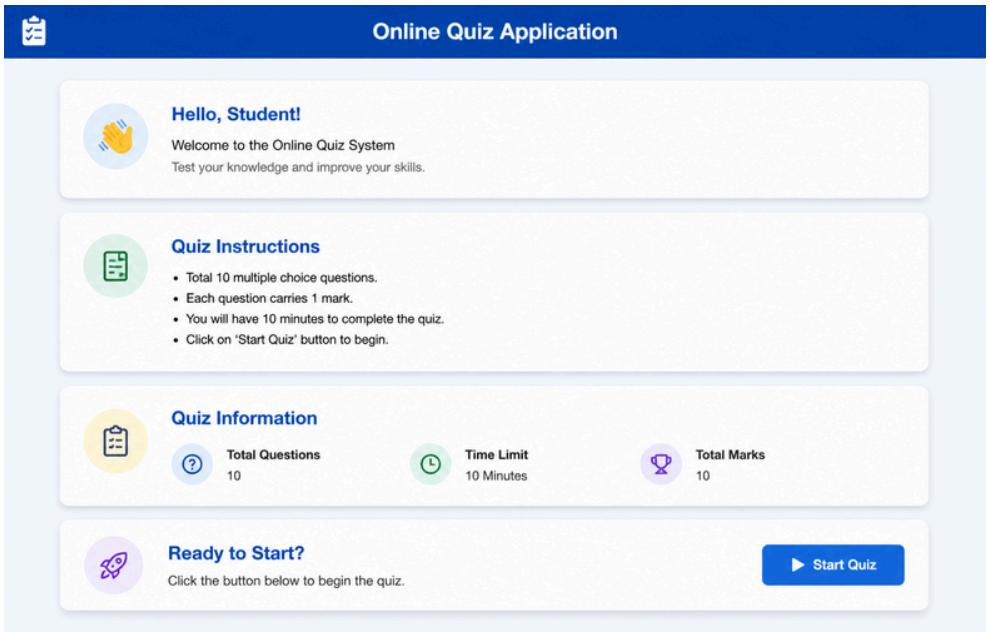
QuizForm.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Quiz</title>
  <style>
    body {
      font-family: Arial;
      background: linear-gradient(to right, #4facfe, #00f2fe);
    }
    .container {
      width: 400px;
      margin: 80px auto;
      background: white;
      padding: 20px;
      border-radius: 10px;
    }
    h2 {
      text-align: center;
    }
    button {
      width: 100%;
      padding: 10px;
      background: blue;
      color: white;
      border: none;
    }
  </style>
</head>

<body>
<div class="container">
  <h2>Online Quiz</h2>

  <form action="quizServlet" method="post">
    <input type="text" name="name" placeholder="Enter Name" required><br><br>

    <p>1. HTML stands for?</p>
    <input type="radio" name="q1" value="a"> Hyper Text Markup Language<br>
    <input type="radio" name="q1" value="b"> High Text Machine Language<br><br>

    <p>2. Java is?</p>
    <input type="radio" name="q2" value="a"> Programming Language<br>
    <input type="radio" name="q2" value="b"> Database<br><br>

    <button type="submit">Submit Quiz</button>
  </form>
</div>
</body>
</html>
```

QuizServlet.java :

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/quizServlet")
public class QuizServlet extends HttpServlet {

   protected void doPost(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();

      String name = request.getParameter("name");
      String q1 = request.getParameter("q1");
      String q2 = request.getParameter("q2");

      int score = 0;

      if ("a".equals(q1)) score++;
      if ("a".equals(q2)) score++;

      out.println("<html><body style='font-
family:Arial;background:#f2f6ff;'>");
      out.println("<div style='width:400px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");
      out.println("<h2>Quiz Result</h2>");
      out.println("<p><b>Name:</b> " + name + "</p>");
      out.println("<p><b>Your Score:</b> " + score + "/2</p>");
      out.println("</div>");
      out.println("</body></html>");
   }
}
```

output :

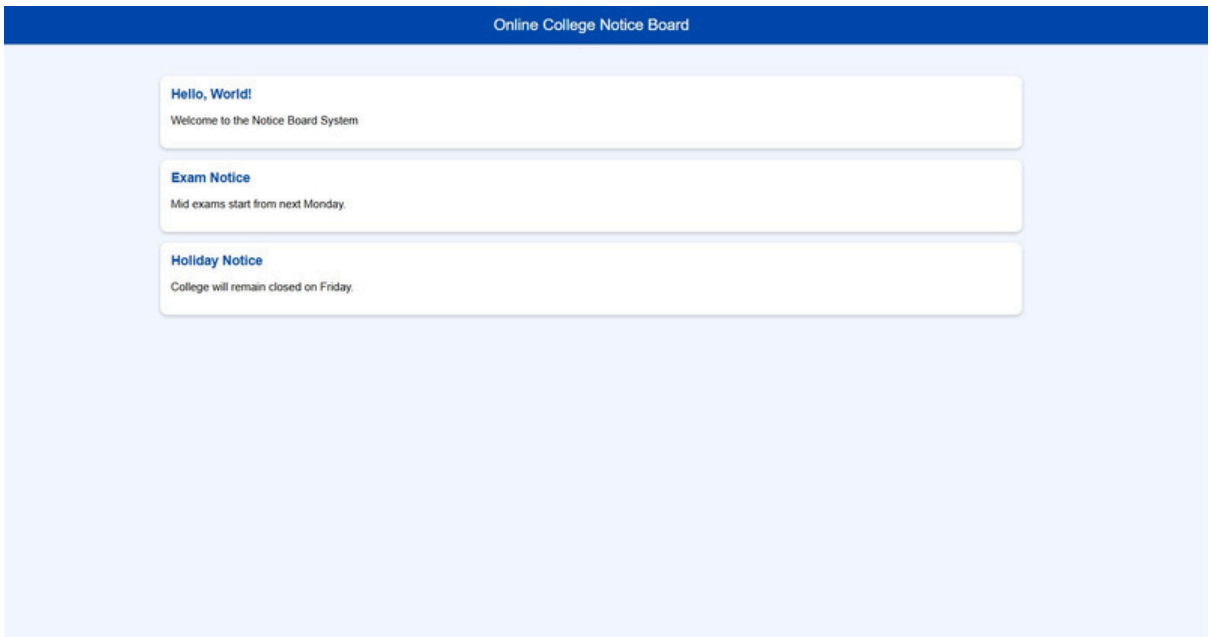


result :

Thus, the Online Quiz Application using Servlet is successfully implemented. The application accepts user input, evaluates quiz answers, and displays the final score through a web browser.

```
    out.println("<div style='width:500px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");
    out.println("<h2 style='color:#004aad;'>Notice Submitted
Successfully</h2>");
    out.println("<p><b>Name:</b> " + name + "</p>");
    out.println("<p><b>Age:</b> " + age + "</p>");
    out.println("<p><b>Department:</b> " + dept + "</p>");
    out.println("<p><b>Roll No:</b> " + roll + "</p>");
    out.println("<p><b>Notice:</b> " + message + "</p>");
    out.println("</div>");
    out.println("</body></html>");
  }
}
```

Output:



RESULT:

The browser will display the message "Hello, World!" returned by the Servlet.